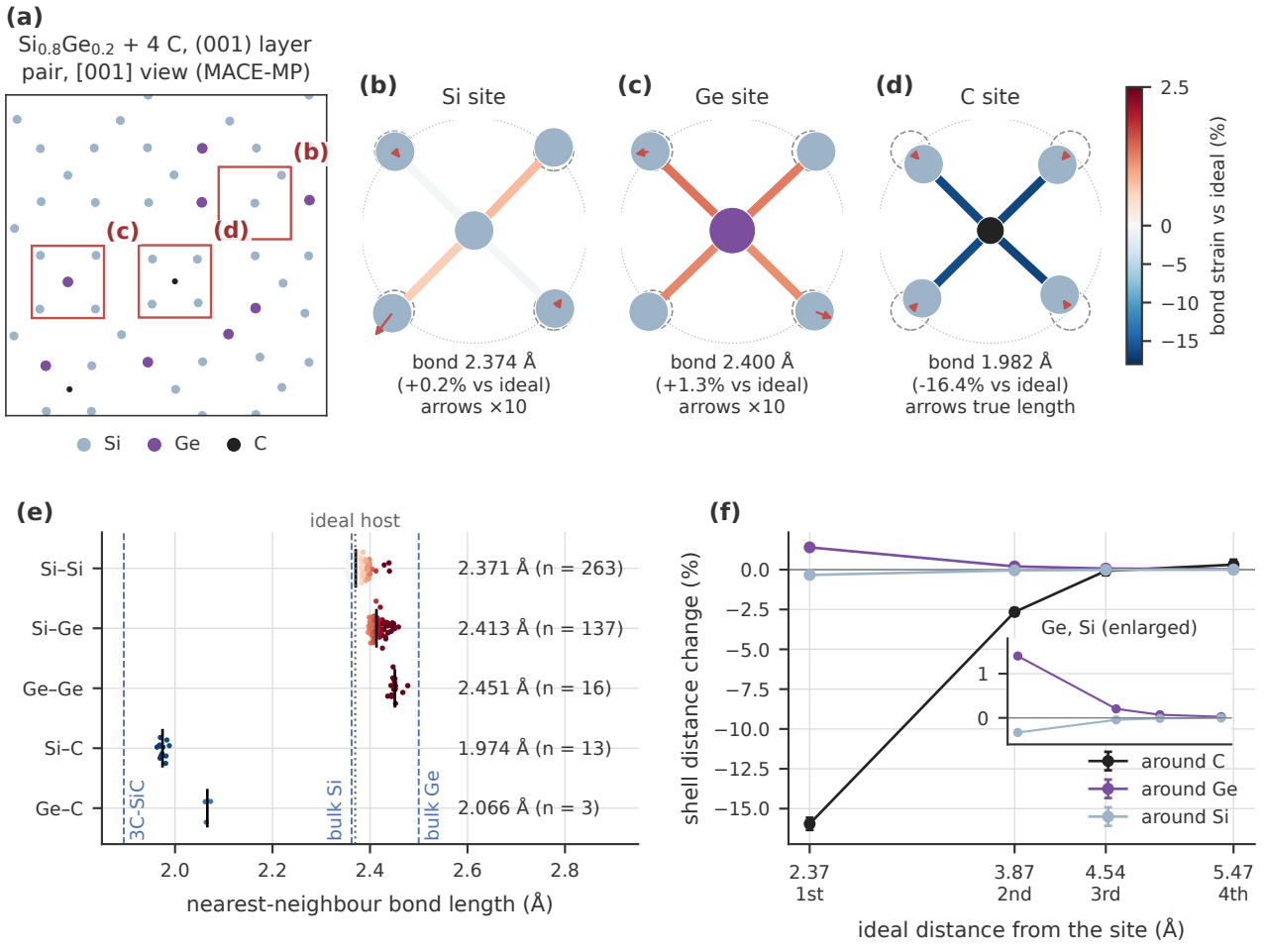

(a)
Si0.8Ge0.2 + 4 C, (001) layer pair, [001] view (MACE-MP)
(b)
(c)
(d)
Si
Ge
C
(b)
Si site
bond 2.374 Å
(+0.2% vs ideal)
arrows ×10
(c)
Ge site
bond 2.400 Å
(+1.3% vs ideal)
arrows ×10
(d)
C site
bond 1.982 Å
(-16.4% vs ideal)
arrows true length
2.5
0
−5
−10
−15
bond strain vs ideal (%)
(e)
ideal host
Si–Si
Si–Ge
Ge–Ge
Si–C
Ge–C
2.371 Å (n = 263)
2.413 Å (n = 137)
2.451 Å (n = 16)
1.974 Å (n = 13)
2.066 Å (n = 3)
3C-SiC
bulk Si
bulk Ge
2.0
2.2
2.4
2.6
2.8
nearest-neighbour bond length (Å)
(f)
shell distance change (%)
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
Ge, Si (enlarged)
1
0
around C
around Ge
around Si
2.37
1st
3.87
2nd
4.54
3rd
5.47
4th
ideal distance from the site (Å)